Lecture 2: PID control

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English version of slides by ©Bo Wahlberg

Outline

Today's Ordlista

English	Svenska
stationary error	stationärt fel
step response	stegsvar
Proportional Integral Derivative	Propotionell Integrerande Deriverande
settling time / transient response	insvängningstid, lösningstid
Integrator windup	Integratoruppvridning
Taylor expansion	Taylorutveckling

Recap

In the previous lecture..

- Dynamical systems = Differential equations
- Transfer function, poles & zeros, stability
- Closed-loop system
- Block diagram representation

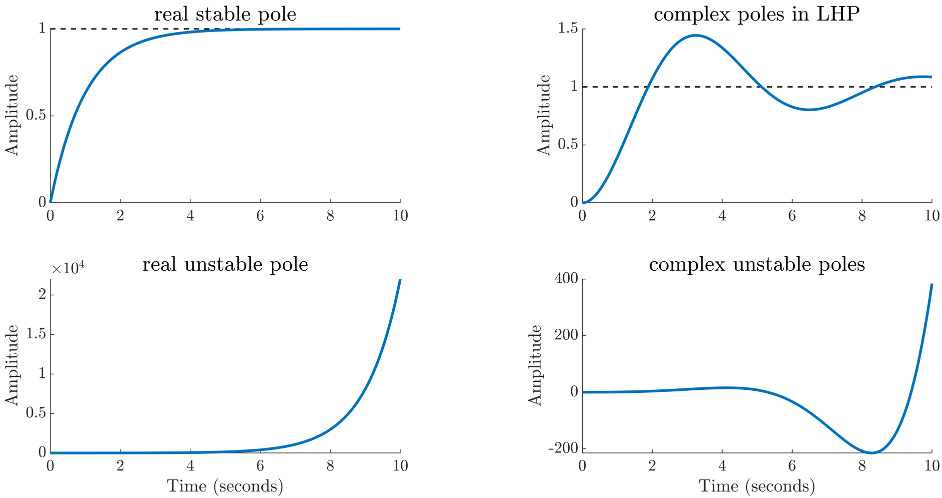
Today:

- PID control

Recap: Transfer function, Poles & Zeros, Stability

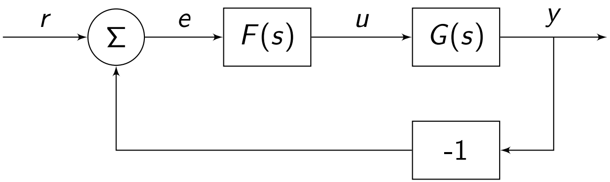
The plots depict the **step response** of 1st (left) and 2nd (right) order systems

system: $u \mapsto y$. step response $= y(t)$ when $u(t) =$ step



Homework: study p. 35-37, Figs. 2.4–2.7, AND Fig. 3.17!

Recap: Block diagrams



$$\implies Y(s) = \underbrace{\frac{G(s)F(s)}{1 + G(s)F(s)}}_{G_c(s)} R(s)$$

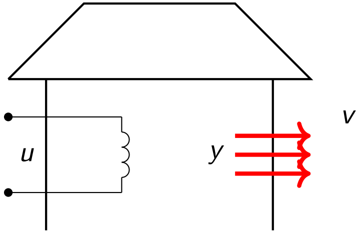
$G_c(s)$ is the closed-loop system's transfer function
Characteristic equation: $1 + G(s)F(s) = 0$

Homework: study pp. 60-63! and try to derive all transfer functions

Example: Temperature control (from Fö1)

How do we control the temperature in a house?

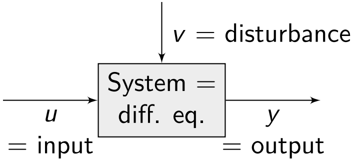
$$\begin{cases} y & = \text{indoor temperature} \\ u & = \text{power supplied to the system} \\ v & = \text{outside temperature} \\ r & = \text{desired temperature} \end{cases}$$



Model: $\dot{y}(t) + \alpha y(t) = \beta u(t) + \alpha v(t)$

Simplified using $\beta = 1$ (scaling):

$$\dot{y}(t) + \alpha y(t) = u(t) + \alpha v(t)$$



How should one determine α? Experiments (another course..)

Open-loop control.. and its problem

Assume that $r = 20°\text{C}$. How should we choose $u(t)$?

Solution:

I. Assume that $\alpha = 1$ and $v = 0$. Choose: $u(t) = \begin{cases} 20, & t \geq 0 \\ 0, & t < 0 \end{cases}$

$$\Rightarrow \dot{y} + y = 20,\ y(0) = 0 \quad \Rightarrow \quad y(t) = 20(1 - e^{-t}) \xrightarrow{t\to\infty} 20 \qquad \textbf{OK!}$$

II. Choose $u(t)$ as in **I** but assume that $\alpha \neq 1$ and $v = -10\,°C$.

$$\Rightarrow \dot{y} + \alpha y = 20 - 10\alpha,\ y(0) = -10$$

$$\Rightarrow \quad y(t) = \frac{20}{\alpha}(1 - e^{-\alpha t}) - 10 \xrightarrow{t\to\infty} \frac{20}{\alpha} - 10$$

Which is **not OK** (unless $\alpha = 2/3$)!

..instead: Use feedback

Model:

$$\begin{cases} y(t) & : \text{indoor temperature} \\ u(t) & : \text{electrical power} \\ v(t) & : \text{outside temperature} \\ r(t) = 20\,^{\circ}C & : \text{desired temperature} \end{cases} \implies \dot{y}(t) + \alpha y(t) = u(t) + \alpha v(t)$$

Goal: To have an $r(t)$ degrees indoor.

Strategy:

- Increase $u(t)$ if it is too cold inside
- Decrease $u(t)$ if it is too warm inside

How much?

Proportional (P) control

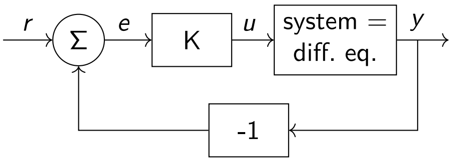
Adjust the control signal proportionally to the size of the error!

$$u(t) = K\left(r(t) - y(t)\right) = K\,e(t)$$

where $K > 0$ is a constant.

Negative feedback!

- Increase $u(t)$ if it is too cold inside:
  $y < r \;\Rightarrow\; e > 0$
- Decrease $u(t)$ if it is too warm inside:
  $y > r \;\Rightarrow\; e < 0$



Even easier strategy: Compare with a thermostat (on/off, switching) (ex: fridge)

P control: Solution

Solution:

- Use $u = K\,e = K\,(r - y)$ in the diff. eq. $\dot{y} + \alpha y = u + \alpha v$:

$$\Rightarrow \dot{y} + \alpha y = Kr - Ky + \alpha v \quad \Rightarrow \quad \dot{y} + (\alpha + K)y = Kr + \alpha v$$

- Let $r = 20$, $v = -10$, and $y(0) = 0$. Then:

$$y(t) = \frac{20\,K}{\alpha + K}\left(1 - e^{-(\alpha+K)t}\right) + \frac{-10\,\alpha}{\alpha + K}\left(1 - e^{-(\alpha+K)t}\right)$$
$$\xrightarrow{t\to\infty} \frac{20}{1 + \frac{\alpha}{K}} - \frac{10}{1 + \frac{K}{\alpha}} \approx 20 \quad \text{(if } K \text{ is big, even though } \alpha \neq 1 \text{ and } v = -10\text{)}$$

Great!, but **stationary error**: $u = Ke > 0$ is required to maintain the correct temperature (otherwise the radiator is closed)

Note that:

- Stationary = "when it has settled (när den har svängt in sig")
- Transient = "while it is still in motion (medan den svänger in sig)"

Why feedback?

:)

- To reduce the impact of disturbances and model error (obtain good control performance despite them)
- To increase speed of transient response/convergence toward steady state Here from $e^{-\alpha t}$ to $e^{-(\alpha+K)t}$.
- To stabilize unstable systems (ex: Segway/inverted pendulum, JAS Gripen)

:(What can go wrong?

- The system does not behave as expected (e.g., fel modell, fel dynamik)
- Limitations in controllability (e.g., elementet har en max effekt)
- Feedback can create instability

Outline

PID control

PID (Proportional Integral Derivative) control solves many control problems.

Example

An industrial process has more than 1000 PID controls

$$\xrightarrow{\;e\;} \boxed{\text{PID}} \xrightarrow{\;u\;}$$

Definition (PID Control)

$$u(t) = K\left(e(t) + \frac{1}{T_I}\int_{t_0}^{t} e(\tau)\,\mathrm{d}\tau + T_D\frac{\mathrm{d}e(t)}{\mathrm{d}t}\right)$$

alt.,

$$u(t) = K_P\,e(t) + K_I\int_{t_0}^{t} e(\tau)\,\mathrm{d}\tau + K_D\,\frac{\mathrm{d}e(t)}{\mathrm{d}t}$$

PID control

It has three terms (three-term controller) that we can fine-tune.

- ***P*roportional feedback**: $K\, e(t)$ is proportional to the current value of the error (the error right now)
- ***I*ntegral feedback**: $\frac{1}{T_I}\int_{t_0}^{t} e(\tau)\,\mathrm{d}\tau$ accounts for past values of the error (how the error *has* behaved)
- ***D*erivative feedback**: $T_D \frac{\mathrm{d}e(t)}{\mathrm{d}t}$ accounts for how the error *will* behave

How should one set up/design K, T_I, and T_D?

$$\Longrightarrow \text{LAB 1!}$$

PI control

$$u(t) = K\Big(e(t) + \frac{1}{T_I}\int_{t_0}^{t} e(\tau)\,\mathrm{d}\tau\Big)$$

The stationary state is $e(t) = 0$, otherwise $u(t)$ increases/decreases due to the integral.

Transient response: Assume that $u = \bar{u}$ is required for $e(t) = 0$. Study

$$\bar{u} = K\Big(e(t) + \frac{1}{T_I}\int_{t_0}^{t} e(\tau)\,\mathrm{d}\tau\Big)$$

If we derive this expression, we obtain

$$K\Big(\dot{e} + \frac{1}{T_I}e\Big) = 0 \quad \Rightarrow \quad e(t) = C\,e^{-t/T_I}$$

If T_I is small, we obtain a faster settling time towards the reference value
(*but may compromise the stability!*, see LABS)

Integrator windup

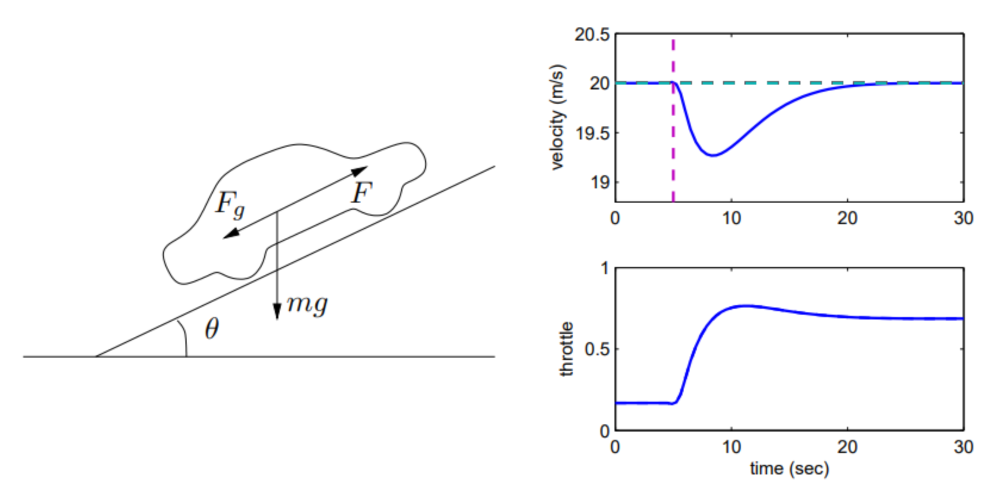
Effects of the saturation: Suppose there is a maximum permissible control signal $u_{\max}$, i.e., $u(t) \leq u_{\max}$. If we use a PID controller, we may have

$$u(t) = K\Big(e(t) + \frac{1}{T_I}\int_{t_0}^{t} e(\tau)\,\mathrm{d}\tau\Big) > u_{\max}$$

- JAS 39 Gripen, 1993, Långholmen (YouTube)
- JAS 39 Gripen, 1989, Pilot Induced Oscillation. (YouTube)"Piloten för snabb P-reglering" <jmfr bilkärning, cykel>
- Boeing 737 Max, 2018, 2019. Maneuvering Characteristics Augmentation System (MCAS): konflikt mellan pilot och nytt reglersystem pga större motor

Integrator windup: An Example

Example (Car in a uphill slope)



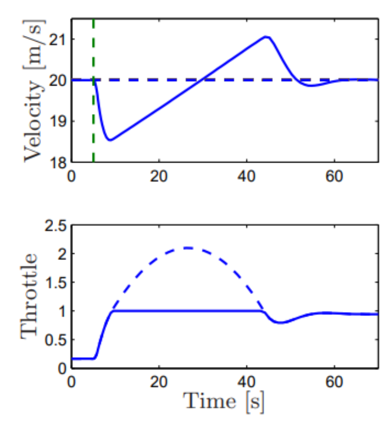
Example on *Cruise Control*, Ch 3.1 & Example 10.2 in "Feedback Systems [..]", by Åström & Murray

Integrator windup: An Example

Example (Car in a uphill slope)

Integrator windup:

- Once the input saturates, the integral of the error keeps increasing
- When the error decreases, the large integral value prevents the controller from resuming "normal operations" quickly (the integral error must decrease first..), hence the response is delayed!

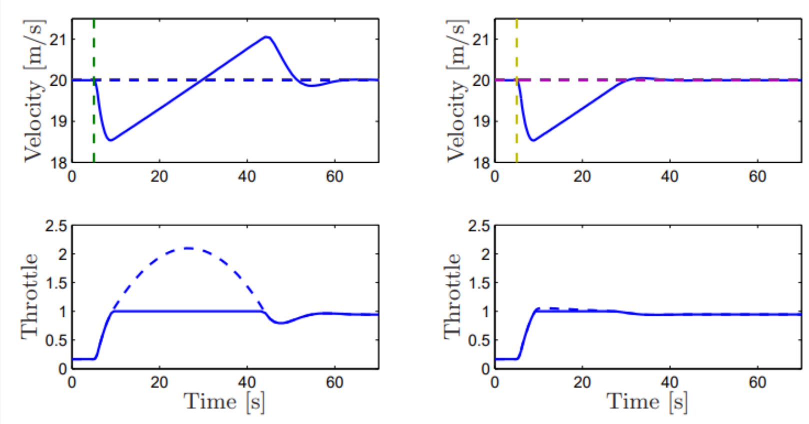


Example on *Cruise Control*, Ch 3.1 & Example 10.2 in "Feedback Systems [..]", by Åström & Murray

Integrator windup: An Example

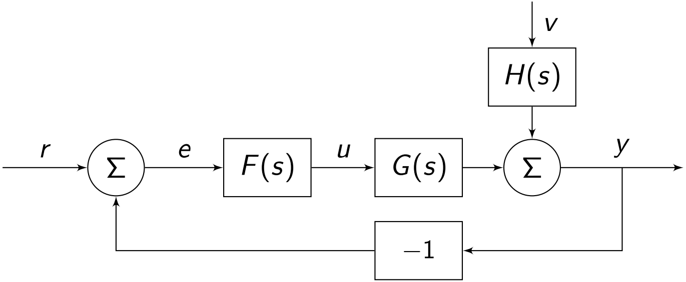
Example (Car in a uphill slope)

Anti-windup (idea): Once the input saturates, stop integrating the error



Example on *Cruise Control*, Ch 3.1 & Example 10.2 in "Feedback Systems [..]", by Åström & Murray

Feedback with the help of a block diagram



$$Y(s) = \frac{G(s)F(s)}{1+G(s)F(s)}R(s) + \frac{H(s)}{1+G(s)F(s)}V(s)$$

PI control: $u(t) = K\left(e(t) + \frac{1}{T_I}\int_{t_0}^{t} e(\tau)\,\mathrm{d}\tau\right)$

$$U(s) = K\left(1 + \frac{1}{T_I}\frac{1}{s}\right)E(s) \quad \Rightarrow \quad F_{\mathrm{PI}}(s) = K\left(1 + \frac{1}{T_I}\frac{1}{s}\right)$$

$H(s)$: without feedback. $1+G(s)F(s)$: feedback

Example: PI control of Temperature

$$G(s) = \frac{1}{s+\alpha}, \quad H(s) = \frac{\alpha}{s+\alpha}$$

$$Y(s) = \frac{K(1+T_I s)}{T_I s^2 + T_I(\alpha+K)s + K} R(s) + \frac{\alpha T_I s}{T_I s^2 + T_I(\alpha+K)s + K} V(s)$$

Stable? Poles in RHP?

$$s^2 + (\alpha + K)s + K/T_I$$

We can choose K and T_I so to obtain arbitrary poles in LHP.

Ex: Poles in -1: $(s+1)^2 = s^2 + 2s + 1 \Rightarrow \begin{cases} K = 2 - \alpha \\ T_I = K = 2 - \alpha \end{cases}$

Example: PI control of Temperature

Assume $r(t) = 20 \;\Rightarrow R(s) = \frac{20}{s}$, and $v(t) = -10 \;\Rightarrow V(s) = \frac{-10}{s}$.

Stationary error?

Solution: Use the final value theorem (Slutvårdes-satsen):

- Assume stability (poles in LHP). Hence, there exists $\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$
- Calculate $Y(s)$ and $sY(s)$:

$$Y(s) = \frac{K(1+T_I s)}{T_I s^2 + T_I(\alpha + K)s + K} R(s) + \frac{\alpha T_I s}{T_I s^2 + T_I(\alpha + K)s + K} V(s)$$

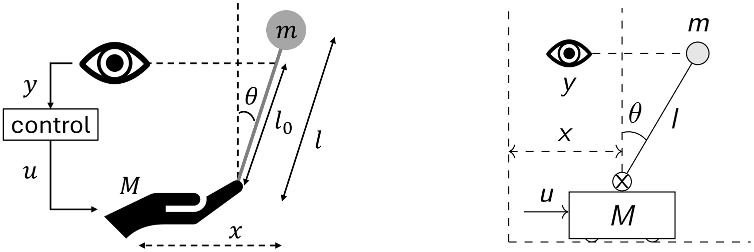
$$\Rightarrow \quad sY(s) = \frac{K(1+T_I s)}{T_I s^2 + T_I(\alpha + K)s + K} 20 + \frac{\alpha T_I s}{T_I s^2 + T_I(\alpha + K)s + K} (-10)$$

Therefore: $\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = 20 + 0$

No stationary error regardless of the outside temperature v!

Inverted pendulum

Examples of balance systems: Segway; Balancing an inverted pendulum on one's palm

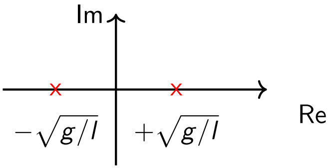


Inverted pendulum: Stability

The system has a transfer function $\approx$ in the form

$$G(s) = \frac{1}{s^2 - g/l}$$

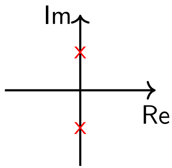
where l is the length of the pendulum.
Obs: unstable system (poles in $\pm\sqrt{g/l}$)!



- **P control**: The closed-loop system's poles are the roots of $s^2 - g/l + K = 0$:

$$G_c(s) = \frac{\frac{K}{s^2-g/l}}{1 + \frac{K}{s^2-g/l}} = \frac{K}{s^2 - g/l + K}$$

Obs: oscillates (when K is big)



- **PI control**: Does not work

In this example, the location of the base does not need to be controlled.

Inverted pendulum: PD control

PD control:

The controller is defined as $F_{\mathrm{PD}}(s) = K(1 + T_D s)$ ("velocity feedback")

$$G_c(s) = \frac{G(s)F(s)}{1 + G(s)F(s)} = \frac{K(1 + T_D s)}{s^2 + KT_D s - g/l + K}$$

The closed-loop system poles are given by

$$s^2 + K\,T_D s - g/l + K = 0$$

Example

Choose: $T_D = 2/K, \;\; K = 1 + g/l$

$\Longrightarrow \quad (s+1)^2 = 0 \quad$ i.e., poles in -1: OK!

Inverted pendulum: PD control

The derivative action speeds up the system and improves stability.

However, if T_D is too large, it has the opposite effect.

If we perform a first-order Taylor expansion of the error:

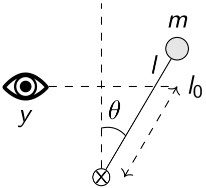
$$e(t + T_D) \approx e(t) + T_D \frac{\mathrm{d}}{\mathrm{d}t} e(t)$$

then we can interpret $T_D \approx$ prediction horizon.

Inverted pendulum

Assume that you can only observe the pendulum up until a certain length $l_0 \leq l$

$\implies$ Unstable zero in $g/(l - l_0)$



It is much more difficult to control – Test!

Outline

Next time: Stability and the Nyquist criterion

Homework:

Prepare for LAB 1! = Tune a PID controller

- Read information on Canvas and LAB-PM
- Read chapters 1.5-3.6. In focus: Figs. 2.5–2.7, Figs. 3.6–3.9, Fig. 3.17
- Solve exercise 2.9